Topic 14: Conditional Probability and Bayes' Rule

02-680: Essentials of Mathematics and Statistics

October 26, 2026

1 Basics

We think of conditional probability as a restriction to some subset of our sample space. If we know one thing happened already whats the probability of another conditioned on the thing we know.

A ***probability space*** $(\Omega, \mathcal{A}, p)$ bundles the three things from last topic: the sample space Ω, the event space $\mathcal{A}$, and the probability function p that assigns each event a number. (For discrete Ω the event space is the power set, so you can mostly ignore it for now; it must be named because $p(A \mid B)$ only makes sense for $A, B \in \mathcal{A}$.) The basic rule of conditional probability is the following: for events $A, B \in \mathcal{A}$

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

Using the normalization axiom ($p(\Omega) = 1$) we can show that

$$p(A \mid \Omega) = \frac{p(A \cap \Omega)}{p(\Omega)} = \frac{p(A)}{1}$$

Going back to the example from last topic if $A_{firstheads}$ is defined as before and $A_{allheads} = \{\langle \texttt{Heads}, \texttt{Heads} \rangle\}$, we can say that

$$p(A_{allheads} \mid A_{firstheads}) = \frac{p(A_{allheads} \cap A_{firstheads})}{p(A_{firstheads})} = \frac{1/4}{1/2} = \frac{1}{2}$$

This is based on the fact that

$$A_{allheads} \subseteq A_{firstheads} \implies A_{allheads} \cap A_{firstheads} = A_{allheads}.$$

This can be interpreted as "the probability you end up with both coin flips being heads, given that the first coin was heads."

If we examine the opposite: "the probability the first coin was heads, given that you end up with both coin flips being heads" the logic here still makes sense

$$p(A_{firstheads} \mid A_{allheads}) = \frac{1/4}{1/4} = 1.$$

Another example. Consider the following scenario for people's illnesses:

Flu? (F)	**Cough?** (C)	$p(C/\overline{C} \cap F/\overline{F})$
F	C	0.3
F	$\overline{C}$	0.2
$\overline{F}$	C	0.15
$\overline{F}$	$\overline{C}$	0.35

How do we calculate the probability you have flu given that you have a cough?

$$p(F \mid C) = \frac{p(C \cap F)}{p(C)}$$

But how do we get $p(C)$?

1.1 Sum Rule.

If we have some partition of the space $A_1, A_2, ..., A_n$ (remember a partition is a set of sets that all are disjoint, but $\bigcup_{i=1}^{n} A_i = \Omega$), then we can say the following:

$$p(B) = \sum_{i=1}^{n} p(B \cap A_i).$$

So, going back to our original question above, we can say that

$$p(C) = p(C \cap F) + p(C \cap \overline{F}) = 0.3 + 0.15$$

Therefore

$$p(F \mid C) = \frac{p(C \cap F)}{p(C)} = \frac{0.3}{0.45} = \frac{2}{3}$$

1.2 Product and Chain Rules.

The product rule, is an algebraic manipulation of the conditional probability definition, but comes in handy quite often (if $p(B) > 0$):

$$p(A \cap B) = p(A \mid B)p(B).$$

The chain rule is an expansion of this to multiple events $D_1, D_2, ...D_n$ (notice there is no condition on these events)

$$
\begin{aligned}
&p(D_1 \cap D_2 \cap ... \cap D_n) \\
&= p(D_n \mid D_1 \cap D_2 \cap ... \cap D_{n-1}) p(D_1 \cap D_2 \cap ... \cap D_{n-1}) \\
&= p(D_n \mid D_1 \cap D_2 \cap ... \cap D_{n-1}) p(D_{n-1} \mid D_1 \cap D_2 \cap ... \cap D_{n-2}) p(D_1 \cap D_2 \cap ... \cap D_{n-2}) \\
&= ... \\
&= p(D_n | D_1 \cap D_2 \cap ... \cap D_{n-1}) ... p(D_3 | D_1 \cap D_2) p(D_2 | D_1) p(D_1) \\
&= p(D_1) p(D_2 | D_1) p(D_3 | D_1 \cap D_2) ... p(D_n | D_1 \cap D_2 \cap ... \cap D_{n-1})
\end{aligned}
$$

1.3 Independence

Events A and B are ***independent*** if

$$p(A \cap B) = p(A) \cdot p(B).$$

They are ***conditionally independent*** if

$$p(A \cap B | C) = p(A|C) \cdot p(B|C).$$

Example. A box contains two coins: a regular coin ($p(\texttt{Heads}) = p(\texttt{Tails}) = \frac{1}{2}$) and one fake two-headed coin ($p(\texttt{Heads}) = 1$), we choose a coin and flip it twice.

Define the following events.

A — First coin toss results in a `Heads` (let $\overline{A}$ be `Tails`)

B — Second coin toss results in a `Heads` (let $\overline{B}$ be `Tails`)

C — the regular coin has been selected (let $\overline{C}$ be the biased coin)

So, the conditional probabilities are as follows:

$$
\begin{aligned}
p(A|C) &= \frac{1}{2} & p(A|\overline{C}) &= 1 \\
p(B|C) &= \frac{1}{2} & p(B|\overline{C}) &= 1 \\
p(A \cap B|C) &= \frac{1}{4} & p(A \cap B|\overline{C}) &= 1 \\
p(C) &= p(\overline{C}) = \frac{1}{2}
\end{aligned}
$$

Since

$$p(A \cap B|C) = \frac{1}{4} = p(A|C) \cdot (B|C)$$

and

$$p(A \cap B|\overline{C}) = 1 = p(A|\overline{C}) \cdot (B|\overline{C})$$

we can see A and B are *conditionally* independent, but are they (regular) independent?

We know from the sum rule that if we have a partition (C and $\overline{C}$) we can say that

$$p(A) = p(A \cap C) + p(A \cap \overline{C})$$

then using the product rule,

$$= p(A|C)p(C) + p(A|\overline{C})p(\overline{C}) = \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{3}{4}$$

$p(B) = \frac{3}{4}$ using the same logic.

But to determine if A and B are independent we need to know $p(A \cap B)$, we can use the same trick:

$$\begin{aligned} p(A \cap B) &= p(A \cap B|C)p(C) + p(A \cap B|\overline{C})p(\overline{C}) \\ &= p(A|C)p(B|C)p(C) + p(A|\overline{C})p(B|\overline{C})p(\overline{C}) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot 1 \cdot \frac{1}{2} \\ &= \frac{1}{8} + \frac{1}{2} = \frac{5}{8} = \frac{10}{16}. \end{aligned}$$

But since $p(A) \cdot p(B) = \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$, we can see the coin flips are not independent.

Example. Consider again rolling a (6-sided) die and let

$A = \{1, 2\}$

$B = \{2, 4, 6\}$

$C = \{1, 4\}$

$$p(A) = \frac{1}{3} \;\; p(B) = \frac{1}{2}$$

$$p(A \cap B) = \frac{1}{6} = p(A) \cdot p(B)$$

But,

$$p(A \mid C) = \frac{1}{2} \;\; p(B \mid C) = \frac{1}{2}$$

$$p(A \cap B \mid C) = p(\{2\} \mid C) = 0$$

thus

$$p(A \cap B \mid C) = 0 \neq \frac{1}{4} = p(A \mid C) \cdot p(B \mid C).$$

2 Bayes' Rule

For some probability space $(\Omega, \mathcal{A}, p)$, and for some event that already happened $B \in \mathcal{A}$. Then the probability that some event $A \in \mathcal{A}$ also happens:

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

(where $p(B) > 0$).

Bayes' Rule * tells us

$$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B)}$$

we call

$p(A)$ — the prior

$p(B \mid A)$ — the likelihood

$p(A \mid B)$ — the posterior

Another form:

$$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B)} = \frac{p(B \mid A)p(A)}{p(B \mid A)p(A) + p(B \mid \overline{A})p(\overline{A})}$$

Example. Let A be event where you have the flu, and B be the event you cough. Given the following information:

Prior: $p(A) = 0.05$ (background probability that you have the flu regardless of symptoms)

*originally published in 1763 by Thomas Bayes

Likelihood:

- $p(B \mid A) = 0.8$ (probability you have a cough given you have the flu),
- $p(B \mid \overline{A}) = 0.2$ (probability you have a cough if you don't have the flu)

Whats the probability you have the flu if you have a cough ($p(A \mid B)$)?

$$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B \mid A)p(A) + p(B \mid \overline{A})p(\overline{A})} = \frac{(0.8)(0.05)}{(0.8)(0.05) + (0.2)(0.95)} = \frac{0.04}{0.04 + 0.19} \approx 0.174$$

What if you're really susceptible to a cough, so $p(A) = 0.5$:

$$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B \mid A)p(A) + p(B \mid \overline{A})p(\overline{A})} = \frac{(0.8)(0.5)}{(0.8)(0.5) + (0.2)(0.5)} = \frac{0.4}{0.4 + 0.1} = 0.8$$

Useful References

Wasserman. "All of Statistics: A Concise Course in Statistical Inference" §1 Degroot and Schervish. "Probability and Statistics" §2